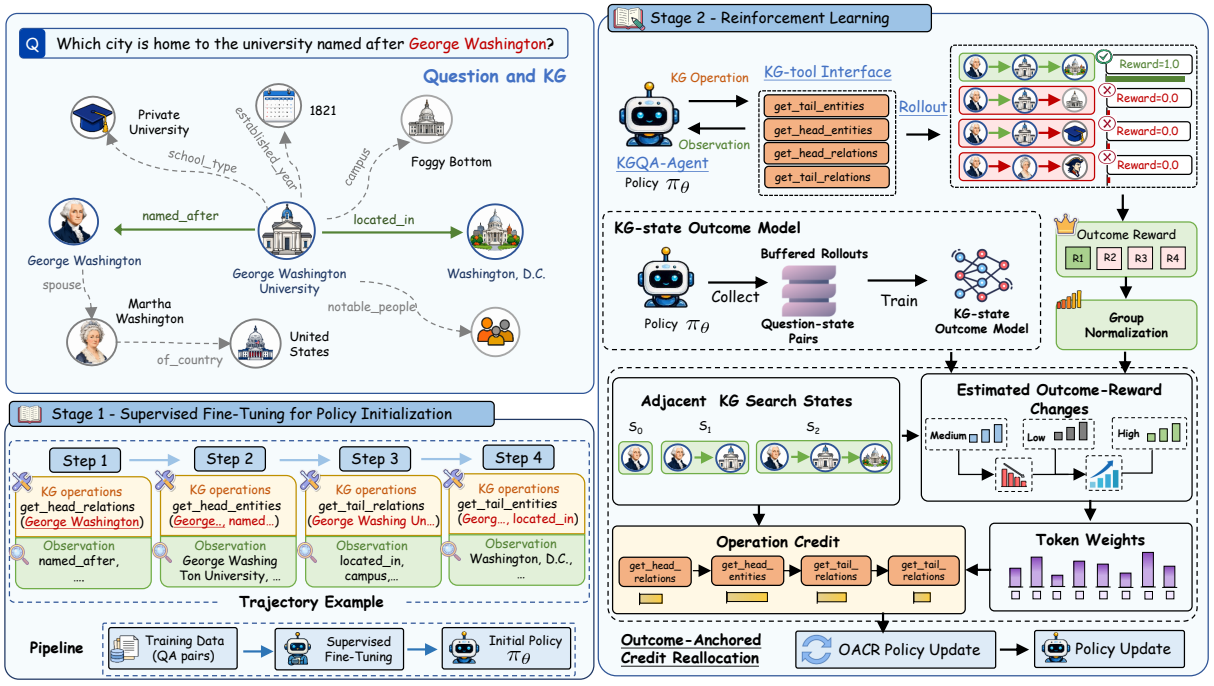

Q
Which city is home to the university named after George Washington?
Question and KG
Private University
1821
established_year
school_type
campus
Foggy Bottom
named_after
located_in
George Washington
George Washington University
Washington, D.C.
spouse
Martha Washington
notable_people
of_country
United States
Stage 1 - Supervised Fine-Tuning for Policy Initialization
Step 1
Step 2
Step 3
Step 4
KG operations
get_head_relations (George Washington)
KG operations
get_head_entities (George.., named...)
KG operations
get_tail_relations (George Washing Un...)
KG operations
get_tail_entities (Georg..., located_in)
Observation
named_after, ...
Observation
George Washing Ton University, ...
Observation
located_in, campus,...
Observation
Washington, D.C., ...
Trajectory Example
Pipeline
Training Data (QA pairs)
Supervised Fine-Tuning
Initial Policy π_θ
Stage 2 - Reinforcement Learning
KG Operation
Observation
KGQA-Agent
Policy π_θ
KG-tool Interface
get_tail_entities
get_head_entities
get_head_relations
get_tail_relations
Rollout
Reward=1.0
Reward=0.0
Reward=0.0
Reward=0.0
KG-state Outcome Model
Policy π_θ
Collect
Buffered Rollouts
Question-state Pairs
Train
KG-state Outcome Model
Outcome Reward
R1
R2
R3
R4
Group Normalization
Adjacent KG Search States
S_0
S_1
S_2
Estimated Outcome-Reward Changes
Medium
Low
High
Operation Credit
get_head_ relations
get_head_ entities
get_tail_ relations
get_tail_ relations
Token Weights
Outcome-Anchored Credit Reallocation
OACR Policy Update
Policy Update